

Lesson 3

The mode superimposition approach

1 The mode superimposition approach as a coordinate transformation

In lesson 1 it was shown that, when considering a system with n degrees of freedom (dof), the linearised form of the equations governing the motion of the system takes the form:

$$[m]\ddot{\underline{x}} + [c]\dot{\underline{x}} + [k]\underline{x} = \underline{f}(t) \quad (1.1)$$

where $\underline{x} = \underline{0}$ represents the equilibrium position around which linearisation has been performed.

In lesson 2, the solutions of equation (1.1) for different types of motion (undamped free motion, damped free motion, forced motion) were studied. In that lesson it was shown that, as soon as the complexity of the system increases, the handling of these solutions becomes increasingly difficult, and the use of numerical methods implemented in a computer code cannot be avoided.

For instance, determining the forced motion of the system involves the solution of a complex-valued linear problem with the same dimension n as the number of system's degrees of freedom. Finding an analytical solution for this kind of problem turns out to be difficult even for a system with two degrees of freedom and becomes totally unpractical for systems with a higher number of degrees of freedom.

On the other hand, solutions found by means of numerical methods provide a very limited insight into the physics of the problem. Therefore, we will introduce in this lesson a new way to describe the motion of a n -dof system, called the “mode superimposition approach”, or simply “modal approach”. By this method, the motion of the system may be interpreted as the superimposition of the motion of n single dof systems: since analytical solutions are known for the motion of a single dof, the modal approach allows to get physical insight about the motion of rather complex systems, by “decomposing” the considered system into a set of single-dof equivalent ones.

In this first section, we start with introducing the mode superimposition approach in the form of a change of coordinates. Given a set of independent coordinates $\underline{x}$ for the system, a different set of coordinates $\bar{\underline{x}}$ may be introduced based on a linear transformation expressed in matrix form as:

$$\underline{x} = [A]\bar{\underline{x}} \quad (1.2)$$

with $[A]$ a square matrix of order n transforming the “new” coordinates of the system $\bar{\underline{x}}$ into the old ones $\underline{x}$. Any change of reference in the form (1.2) may be defined, as long as matrix $[A]$ is non singular: this condition guarantees that the inverse of matrix $[A]$ exists, and thus that equation (1.2) defines a bi-univocal relationship between the old and the new sets of coordinates.

Now, the question arises if there exists any particular change of coordinates that leads to a simplified structure of the equations of motion (1.1). Quite obviously, the best possible form for that equation would be one where the mass, stiffness and damping matrices are diagonal, since in that case the system of n equation of motion could be regarded actually as n decoupled equations each one describing the motion of a different single degree of freedom system. We anticipate here (and will prove later) that such a change of coordinates exists indeed for any multi-dof system, and is obtained using as the matrix $[A]$ of (1.2) the modal matrix $[\Phi]$ introduced at the end of Lesson 2.

The new coordinates defined this way, will be denoted by symbol $\underline{q}$, indicating obviously a vector formed by n elements and will be called “modal coordinates” or “principal coordinates” to underline their property of de-coupling the equations of the system.

Thus, the modal approach is defined by the relationship:

$$\underline{x} = [\Phi] \underline{q} \quad (1.3)$$

we recall, by the way, that in lesson 2 it was observed that the modal matrix is for sure non singular, and thus the change of coordinates defined by equation (1.3) is always possible.

To obtain a new form for the equations of motion of the system, based on the new modal coordinates $\underline{q}$ instead than on the old ones $\underline{x}$, we recall the expression of the kinetic energy T , total potential energy V , damping function D and virtual work δW prior to the application of Lagrange equation in matrix form (see lesson 1):

$$T = \frac{1}{2} \dot{\underline{x}}^T [m] \dot{\underline{x}} \quad ; \quad V = \frac{1}{2} \underline{x}^T [k] \underline{x} \quad ; \quad D = \frac{1}{2} \dot{\underline{x}}^T [c] \dot{\underline{x}} \quad ; \quad \delta W = \delta \underline{x}^T \underline{f} \quad (1.4)$$

where $[m]$, $[k]$ and $[c]$ are constant valued, square matrices.

We replace then in expressions (1.4) the relationship (1.3) between the old and new (modal) coordinates, as well as the time derivative of eq. (1.3). To this end, we recall that the modal matrix is constant with respect to time, and thus:

$$\dot{\underline{x}} = [\Phi] \dot{\underline{q}} \quad (1.3b)$$

By this substitution, the expressions of the quantities used by Lagrange equations become:

$$T = \frac{1}{2} \dot{\underline{q}}^T [\Phi]^T [m] [\Phi] \dot{\underline{q}} \quad ; \quad V = \frac{1}{2} \underline{q}^T [\Phi]^T [k] [\Phi] \underline{q} \quad ; \quad D = \frac{1}{2} \dot{\underline{q}}^T [\Phi]^T [c] [\Phi] \dot{\underline{q}} \quad ; \quad \delta W = \delta \underline{q}^T [\Phi]^T \underline{f} \quad (1.5)$$

finally, by applying Lagrange equation in matrix form to the above expressions, using $\underline{q}$ as the vector of new independent coordinates of the system:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\underline{q}}} \right)^T - \left(\frac{\partial T}{\partial \underline{q}} \right)^T + \left(\frac{\partial D}{\partial \dot{\underline{q}}} \right)^T + \left(\frac{\partial V}{\partial \underline{q}} \right)^T = \frac{\delta W}{\delta \underline{q}}$$

the following system of equation of motion is obtained:

$$\| [\Phi]^T [m] [\Phi] \ddot{\underline{q}} + [\Phi]^T [c] [\Phi] \dot{\underline{q}} + [\Phi]^T [k] [\Phi] \underline{q} = [\Phi]^T \underline{f} \| \quad (1.6)$$

thus, it may be observed that changing the original set of independent coordinates $\underline{x}$ into the new one $\underline{q}$ leads to a new system of equations of motion, where the new mass, stiffness and damping matrices may be derived from the original ones by pre-multiplying each matrix by the transpose of the modal matrix and post-multiplying this product again by the modal matrix. In a similar way, the new vector of Lagrangian components of the time dependent forces at second side of the new set of equations may be derived by pre-multiplying the old one by the transpose of the modal matrix.

Until now, no big step forward has apparently been done with respect to the original system of equations of motion (1.1). However, in the next section it will be shown that, under totally general conditions, the new mass and stiffness matrices appearing in equation (1.6) are diagonal matrices and that under specific assumptions, also the damping matrix will be a diagonal matrix. These observations simplify the analysis of the motion of a n -dof system by decomposing it into the study of n decoupled single-dof systems, an approach that will be fully developed in section 3 of this lesson.

2 Orthogonality of the modes of vibration (NOT DONE IN CLASS!)

Aim of this section is to prove first that the **mass and stiffness matrices** in equation (1.6) **are diagonal matrices**. Then, the conditions upon which also the damping matrix in eq. (1.6) turns out to be diagonal will be briefly discussed.

We start by **defining** $[\bar{m}]$ and $[\bar{k}]$ the **modal mass and stiffness matrices** defined by eq. (1.6):

$$[\bar{m}] = [\Phi]^T [m] [\Phi] \quad ; \quad [\bar{k}] = [\Phi]^T [k] [\Phi] \quad (2.1)$$

we then note that, since the **columns of the modal matrix are the modes of vibration** $\underline{x}_0^{(j)}$ ($j=1,2, \dots, n$) of the system (defined according to the original coordinates $\underline{x}$), the element $\bar{m}_{j\ell}$ in row j and column ℓ of the modal mass matrix $[\bar{m}]$ and the element $\bar{k}_{j\ell}$ in row j and column ℓ of the modal stiffness matrix $[\bar{k}]$ are defined by the following matrix products:

$$\bar{m}_{j\ell} = \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} \quad ; \quad \bar{k}_{j\ell} = \underline{x}_0^{(j)T} [k] \underline{x}_0^{(\ell)} \quad (2.2)$$

now, from the definition of the j -th and ℓ -th modes of vibration of the original system (see Lesson 2), we get:

$$-\omega_j^2 [m] \underline{x}_0^{(j)} + [k] \underline{x}_0^{(j)} = \underline{0} \quad ; \quad -\omega_\ell^2 [m] \underline{x}_0^{(\ell)} + [k] \underline{x}_0^{(\ell)} = \underline{0} \quad (2.3)$$

now we pre-multiply the first of equations (2.3) by the transpose of $\underline{x}_0^{(\ell)}$ and the second of equations (2.3) by the transpose of $\underline{x}_0^{(j)}$ and we get:

$$-\omega_j^2 \underline{x}_0^{(\ell)T} [m] \underline{x}_0^{(j)} + \underline{x}_0^{(\ell)T} [k] \underline{x}_0^{(j)} = 0 \quad ; \quad -\omega_\ell^2 \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} + \underline{x}_0^{(j)T} [k] \underline{x}_0^{(\ell)} = 0 \quad (2.4)$$

finally, we transpose the first of equations (2.4), change the sign and add to the second of equations (2.4). In doing so, we recall that $[m]$ and $[k]$ are symmetric matrices:

$$-\omega_j^2 \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} + \underline{x}_0^{(j)T} [k] \underline{x}_0^{(\ell)} + \omega_\ell^2 \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} - \underline{x}_0^{(j)T} [k] \underline{x}_0^{(\ell)} = (\omega_j^2 - \omega_\ell^2) \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} = 0 \quad (2.5)$$

If we now restrict to the case where all the natural frequencies ω_j ($j=1,2, \dots, n$) of the system are distinct, since $(\omega_j^2 - \omega_\ell^2) \neq 0$ we must conclude that for all $j \neq \ell$:

$$\bar{m}_{j\ell} = \underline{x}_0^{(j)T} [m] \underline{x}_0^{(\ell)} = 0 \quad (2.6)$$

and then, using the above result in the second of equations (2.4), also that for all $j \neq \ell$:

$$\bar{k}_{j\ell} = \underline{x}_0^{(j)T} [k] \underline{x}_0^{(\ell)} = 0 \quad (2.7)$$

The property stated by equations (2.6) and (2.7) are known under the name of “orthogonality of the modes of vibration”. It may be observed that this property was proved under the assumption of the system having distinct natural frequencies. However, a more general (and slightly more complex) proof exists, that allows to conclude that the orthogonality property holds for all systems having equations in the form (1.1).

As far as the damping matrix $[c]$ is concerned, it is not possible to state in general that this matrix will also be diagonalised by the mode superimposition approach. However, **there are special cases in which this property applies**. These cases are particularly relevant, because **all the three matrices in equations (1.6) are then diagonal**, and **thus the system of n equations of motion can be decomposed into n single equations**, representing the motion of one single modal coordinate as a single-dof system.

The most remarkable case when a matrix $[c]$ is transformed into a diagonal matrix by the modal approach is the one called “proportional damping”, which means that the damping matrix $[c]$ is defined as a linear combination of the mass and stiffness matrices, $[m]$ and $[k]$ respectively:

$$[c] = \alpha [m] + \beta [k] \quad (2.8)$$

where α and β are two constant parameters, whose values define the entity of damping actions in the system. We note by the way, that these two parameters have physical dimensions: in the International System, α being expressed in s^{-1} and β in s .

It's anticipated that the “proportional damping” assumption is applied in many cases where the actual damping present in the system is not precisely known, and/or a model for dissipative forces in the system is not available or too complicate and uncertain. This is typically the case of the internal damping of distributed parameter systems. We will come back on this point when treating the topic of finite elements analysis of distributed parameter systems.

For the time being, we just note that since both the mass and stiffness matrices are diagonalised by the modal approach, than a damping matrix defined through equation (2.8) will also be diagonalised:

$$[\Phi]^T [c] [\Phi] = \alpha [\Phi]^T [m] [\Phi] + \beta [\Phi]^T [k] [\Phi] = \alpha [\bar{m}] + \beta [\bar{k}] \quad (2.9)$$

Since in this case the n equations of motion of the system are decoupled, we may consider each single equation as representing a single dof system:

$$\bar{m}_i \ddot{q}_i + \bar{c}_i \dot{q}_i + \bar{k}_i q_i = \bar{f}_i(t) \quad (2.10)$$

where $\bar{m}_i$, $\bar{c}_i$ and $\bar{k}_i$ (called respectively the modal mass, modal damping and modal stiffness of the i -th mode of vibration) represent the diagonal elements in the i -th row of the mass, damping and stiffness matrices respectively, and $\bar{f}_i$ is the element in the i -th row of vector:

$$\bar{f} = [\Phi]^T f \quad (2.11)$$

Now, the fact that the equations of motion of the system are decoupled by the modal coordinates q suggests an approach to studying the motion of the system based on the following steps:

1. given the system's equations in the physical coordinates x , which means in the form (1.1), compute the natural frequencies and modes of vibration of the system and form the modal matrix $[\Phi]$;
2. transform the system of equations (1.1) into n decoupled 1-dof equations in the form (2.10), using the modal approach (under the assumption that the damping matrix $[c]$ is also transformed into a diagonal matrix);
3. solve separately the n 1-dof equations for the modal coordinates q_i : this step may take advantage from the fact that analytical solutions exist for the motion of the single-dof system, see Lesson 2;
4. once known the motion of the system in terms of modal coordinates q_i , the motion in terms of physical variables x_i may be obtained by using the mode superimposition equation (1.3). This last step is required since the motion of the system is more clearly described in terms of physical coordinates representing the displacement of notable points like the bodies' centre of mass, or rotations of the bodies. These quantities are more directly related to a physical interpretation of the system's behaviour than the modal coordinates and thus are better suited to represent the motion of the system.

In the next section the above solution strategy is applied to the simple 2-dof system already introduced in the previous lesson.

Before closing this section, we come back to equation (1.3) and, by recalling that the columns of matrix $[\Phi]$ are the system's modes of vibration, we expand this equation into the following sum:

$$\underline{x} = [\Phi] \underline{q} = x_0^{(1)} q_1 + x_0^{(2)} q_2 + \dots + x_0^{(n)} q_n \quad (2.12)$$

where we see that applying the mode superimposition means expanding the motion of the system into a linear combination of motion components, each i -th one having the shape of the i -th mode of vibration $x_0^{(i)}$ and amplitude defined by the i -th modal coordinate q_i . The fact that using the modal coordinates $\underline{q}$ instead than the physical ones $\underline{x}$ leads to a decoupled system of equations means that these “elementary” motion components may be treated separately and then superimposed without the need to consider any “cross-influence” effect between them.

For instance, if we are considering the system subjected to harmonic excitation with frequency Ω , we may easily study the system's resonance conditions through equations (2.10) and (2.12) by recognising that the modal coordinates whose resonance frequency is closer to the external excitation frequency Ω experience the higher amplitudes of vibration and are hence called the “resonant modes” of the system. If the resonances of the system are well separated one from the other, then for a given excitation frequency Ω the system will have only one resonant mode, and if the system is low damped, that the amplitude of the resonant modal coordinate will be much larger compared to the other modes that its contribution in the linear combination of eq. (2.12) will become predominant. We see then that in this resonance condition the motion of the system will have the same shape as the single resonant mode of vibration. Furthermore, depending on the particular shape of the resonant mode, we may immediately identify the physical coordinates of the system that will undergo the larger amplitudes of vibration; these will correspond to the elements of the resonant mode of vibration vector characterised by the higher absolute values. A further consequence of the above considerations is that if a “nodal point” (which means, a point where the value of the mode of vibration is zero) exists for the resonant mode, no resonance peak will appear on that coordinate.

From the above considerations, we see immediately that the modal approach allows us to understand clearly what the vibration of the system will be and where the larger amplitudes of vibration (or strain or whatever mechanical quantity of interest) will take place, depending on the frequency of excitation.

In order to achieve a full understanding of the system's behaviour, the above considerations must be combined with the knowledge of how each modal coordinate will be excited by physical forces acting on the system. To this end, we expand the i -th row of equation (2.11) to define the Lagrangian component of time dependent forces along the generic i -th modal coordinate:

$$\bar{f}_i = x_0^{(i)T} \underline{f} = x_{01}^{(i)} f_1 + x_{02}^{(i)} f_2 + \dots + x_{0n}^{(i)} f_n \quad (2.13)$$

we see from this equation that the total Lagrangian component $\bar{f}_i$ along the i -th modal coordinate is given by a linear combination of the n Lagrangian components acting on the physical coordinates $\underline{x}$ (which will often represent physical forces directly applied on the system), where the coefficients of the linear combination are represented by the values of the i -th mode of vibration vector $x_0^{(i)}$ along each physical coordinate x_i . In case the system is excited by a single force, it is clear from Eq. (2.13) that the largest forcing effect on the i -th modal coordinate will be obtained if the force is applied in a point of the system where the maximum amplitude of vibration is achieved in the considered i -th mode; oppositely, if the single force is applied in a point that is nodal for the i -th mode of vibration, the Lagrangian component $\bar{f}_i$ for that mode will be zero, which means that no excitation will be applied on that component of the system's motion. Thus, even if the single time dependent force is acting at the resonance frequency of the i -th mode $\Omega = \omega_i$, no resonance phenomena will take place

because, the force is not able to introduce energy into the i -th modal component of system's motion, due to the special point where it is applied (the node).

3 An example of application of the mode superimposition approach

We conclude Lesson 3 by showing an application of the mode superimposition approach to a simple 2-dof system, the same used in the example of Lesson 2, with a harmonic force applied on the left body, see figure 3.1 below:

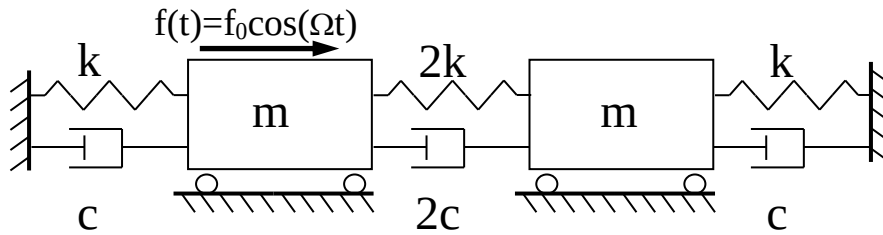


Figure 3.1: 2-dof system formed by two masses moving in the horizontal direction

We take as the physical independent coordinates of the system the horizontal absolute displacement of the left body x_1 and the horizontal absolute displacement of the right body x_2 . By applying Lagrange Equations or any other method, it is an easy task to write the equations of motion of the system, that take the matrix form:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 3c & -2c \\ -2c & 3c \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} + \begin{bmatrix} 3k & -2k \\ -2k & 3k \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} f \\ 0 \end{Bmatrix} \quad (3.1)$$

as we have seen in Lesson 2, the undamped natural frequencies and modes of vibration of the system are respectively the square roots of the eigenvalues and the eigenvectors of matrix:

$$[m]^{-1}[k] = \begin{bmatrix} 3\frac{k}{m} & -2\frac{k}{m} \\ -2\frac{k}{m} & 3\frac{k}{m} \end{bmatrix}$$

which means the solutions of the problem:

$$\left(\omega^2 [I] - [m]^{-1}[k] \right) \underline{x}_0 = \begin{bmatrix} \omega^2 - 3\frac{k}{m} & 2\frac{k}{m} \\ 2\frac{k}{m} & \omega^2 - 3\frac{k}{m} \end{bmatrix} \begin{Bmatrix} x_{01} \\ x_{02} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (3.2)$$

now in order to get non-trivial solution, we require:

$$\det \left(\omega^2 [I] - [m]^{-1}[k] \right) = \omega^4 - 6\frac{k}{m}\omega^2 + 5\left(\frac{k}{m}\right)^2 = 0$$

which provides as solutions the two natural circular frequencies of the system:

$$\omega_1 = \sqrt{\frac{k}{m}} \quad ; \quad \omega_2 = \sqrt{\frac{5k}{m}} \quad (3.3)$$

in order to compute the mode of vibration $\underline{x}_0^{(1)}$ associated with the first natural frequency of the system we replace $\omega = \omega_1$ into equation (3.2), and thus we get:

$$\begin{bmatrix} -2\frac{k}{m} & 2\frac{k}{m} \\ 2\frac{k}{m} & -2\frac{k}{m} \end{bmatrix} \begin{Bmatrix} x_{01}^{(1)} \\ x_{02}^{(1)} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

where we see that the second row of the matrix equation repeats the first one (with the sign changed) and may thus be eliminated. Considering then only the first row of the equation, we get:

$$-2\frac{k}{m}x_{01}^{(1)} + 2\frac{k}{m}x_{02}^{(1)} = 0 \quad \rightarrow \quad x_{02}^{(1)} = x_{01}^{(1)}$$

as expected, the above equation is under-determined (1 equation in 2 unknowns) and thus we need to set arbitrarily the value of one of the unknowns. By setting equal to unity the first element of the mode of vibration vector we get:

$$\underline{x}_0^{(1)} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

To compute the mode of vibration $\underline{x}_0^{(2)}$ associated with the second natural frequency the above procedure is repeated replacing this time into eq. (3.2) $\omega = \omega_2$. We get this time:

$$\begin{bmatrix} 2\frac{k}{m} & 2\frac{k}{m} \\ 2\frac{k}{m} & 2\frac{k}{m} \end{bmatrix} \begin{Bmatrix} x_{01}^{(2)} \\ x_{02}^{(2)} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

where the two rows are equal and may then be reduced to the single equation:

$$2\frac{k}{m}x_{01}^{(2)} + 2\frac{k}{m}x_{02}^{(2)} = 0 \quad \rightarrow \quad x_{02}^{(2)} = -x_{01}^{(2)}$$

and, by setting equal to one the first element of the vector, we get:

$$\underline{x}_0^{(2)} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$$

then, the modal matrix $[\Phi]$ for the system is:

$$[\Phi] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Now we use these results to apply the mode superimposition approach for the system. To this end we define the system's modal coordinates q_1, q_2 by the following equation:

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix} \quad (3.4)$$

we then get:

$$[\bar{m}] = [\Phi]^T [m] [\Phi] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2m & 0 \\ 0 & 2m \end{bmatrix}$$

$$[\bar{k}] = [\Phi]^T [k] [\Phi] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2k & 0 \\ 0 & 10k \end{bmatrix}$$

$$[\bar{c}] = [\Phi]^T [c] [\Phi] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3c & -2c \\ -2c & 3c \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2c & 0 \\ 0 & 10c \end{bmatrix}$$

$$\bar{f} = [\Phi]^T f = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{Bmatrix} f \\ 0 \end{Bmatrix} = \begin{Bmatrix} f \\ f \end{Bmatrix}$$

According to the above results, the motion of the system may be described in terms of modal coordinates by the two decoupled 1-dof equations:

$$\left\{ \begin{array}{l} 2m\ddot{q}_1 + 2c\dot{q}_1 + 2kq_1 = f(t) \end{array} \right. \quad (3.5a)$$

$$\left\{ \begin{array}{l} 2m\ddot{q}_2 + 10c\dot{q}_2 + 10kq_2 = f(t) \end{array} \right. \quad (3.5b)$$

To solve these two equations and find the motion of the system, we confine ourselves to the stationary motion of the system after the initial transient which means, from the mathematical point of view, that we consider only the “particular integral” of the solution, neglecting the general integral of the associated homogeneous equation.

Furthermore, for the sake of simplicity we will initially consider the case of the undamped system ($c=0$) and will then generalise the results to the presence of damping¹.

Equations (3.5a,b) with damping neglected and considering a harmonic force acting on the system with Ω as the forcing frequency become:

$$2m\ddot{q}_1 + 2kq_1 = f_0 \cos(\Omega t) \quad (3.6a)$$

$$2m\ddot{q}_2 + 10kq_2 = f_0 \cos(\Omega t) \quad (3.6b)$$

The particular integrals for these two equations take the form:

$$q_1 = q_{01} \cos(\Omega t) \quad ; \quad q_2 = q_{02} \cos(\Omega t) \quad (3.7)$$

and, by substitution we get:

$$q_{01} = \frac{f_0}{2k} \frac{1}{1 - \left(\frac{\Omega}{\omega_1}\right)^2} \quad ; \quad q_{02} = \frac{f_0}{10k} \frac{1}{1 - \left(\frac{\Omega}{\omega_2}\right)^2} \quad (3.8)$$

where use has been made of expressions (3.3). The values of q_{10} and q_{20} (eq. 3.8) define the amplitude and phase of the harmonic response of the undamped system as functions of frequency and are plotted in figure 3.2. Normally, a representation of q_{10} and q_{20} in terms of modulus and phase (the well known *Bode diagram*) is used. However, since in absence of damping these quantities are positive/negative real valued, it is more practical for the time being to use the representation of figure 3.2 based on real values.

¹ It could be cause of confusion the observation that in absence of damping the transient motion represented by the general integral of the associated homogeneous equation should not decay to zero for increasing time, and thus the calculation of the particular integral alone would make no sense. However, we may consider the case where a very light amount of damping may be (as a first approximation) neglected in the calculation of the particular integral, while it will bring down to zero the transient motion after a sufficiently long time.

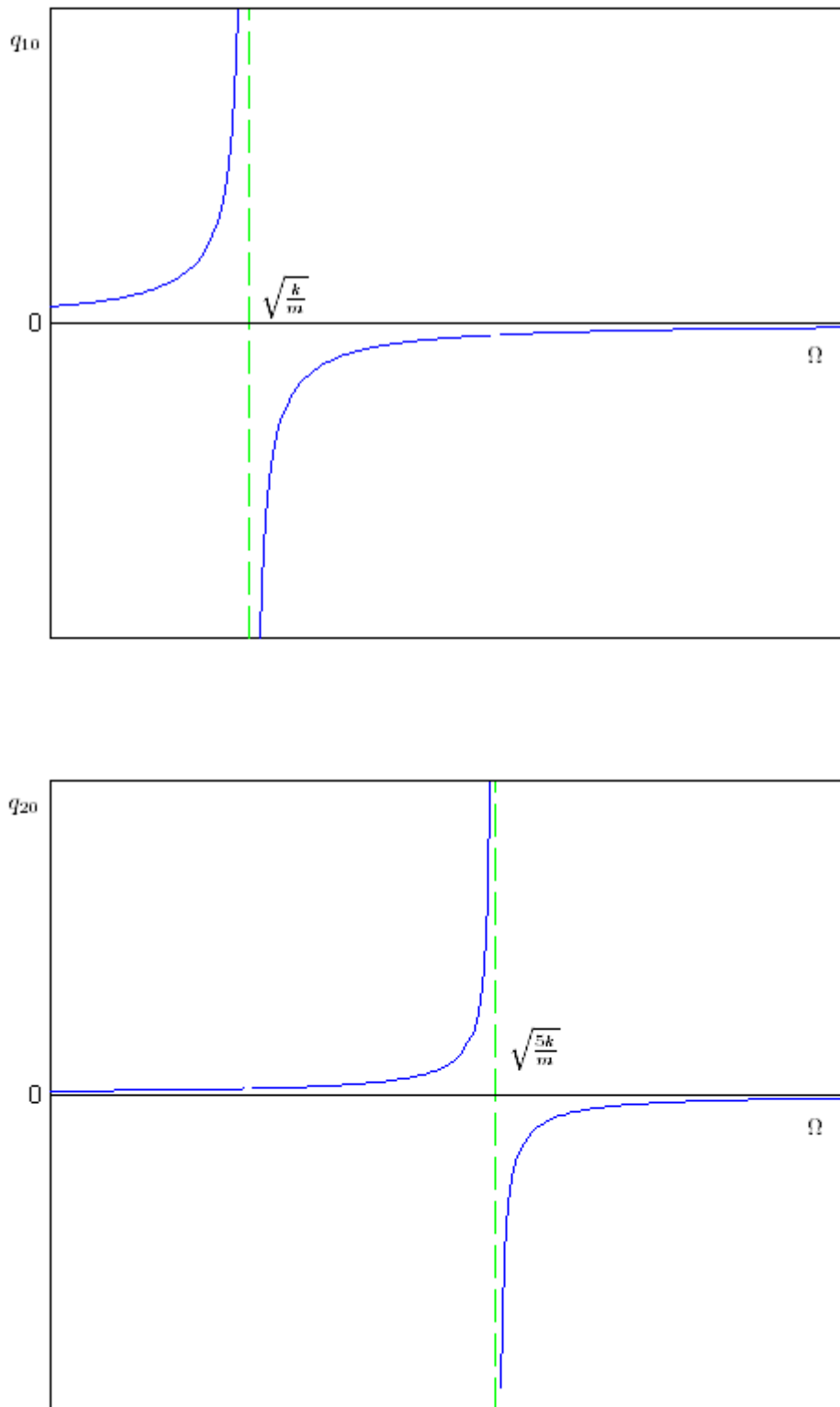


Figure 3.2: Diagrams showing the response of the modal coordinates q_1 and q_2 to harmonic forcing on the left slider, as function of the forcing frequency Ω - undamped case.

We recall that the amplitude and phase of system's vibration in terms of the physical coordinates x_1 and x_2 may be computed using Eq (3.4), i.e. by superimposing the effect of the two modal contributions represented by coordinates q_1 and q_2 .

Expanding the matrix equation (3.4) into two scalar ones brings:

$$\begin{cases} x_1 = q_1 + q_2 \\ x_2 = q_1 - q_2 \end{cases} \quad (3.9)$$

showing that the physical displacements of the left and right slider may be obtained as respectively the sum and the difference of the two modal coordinates. By recalling the expression obtained for the modal coordinates (3.7), the following result is obtained:

$$\begin{cases} x_1 = x_{01} \cos(\Omega t) = (q_{01} + q_{02}) \cos(\Omega t) \\ x_2 = x_{02} \cos(\Omega t) = (q_{01} - q_{02}) \cos(\Omega t) \end{cases} \quad (3.10)$$

The diagrams representing x_{01} and x_{02} as positive/negative real valued are shown in figure 3.3 and are obtained by simply applying eq. (3.10) to the diagrams of figure 3.2 for each value of the forcing frequency Ω . From the graphs in figure 3.3 it is easy to define the amplitude and phase of the vibrations of the sliders, since the absolute value of the plotted quantity represents the amplitude of vibration of the considered body, and a positive sign indicates that the body moves in phase with the applied force (i.e., when the force points rightwards the system is displaced at the right of its equilibrium position and vice-versa), while a negative sign indicates that the body moves in 180° out-of-phase with respect to the force (the displacement of the body happens leftwards when the force is pointing rightwards and vice-versa).

Finally, we may translate the results shown in figure 3.3 into the classical Bode diagrams representation, by simply taking the absolute value of the real-valued diagrams as the amplitude plots of the Bode diagrams and by assigning a zero or $-\pi$ phase depending on the sign: this result is shown in figure 3.4.

We conclude the treatment of the undamped case by reporting some considerations on the physical interpretation of the Bode diagrams in figure 3.4: we observe first of all that two resonance peaks are observed for both slider displacements, corresponding to the two undamped natural frequencies of the system. Since damping is neglected, the amplitude of the resonance peaks tends to infinity when the forcing frequency approaches one of the two natural frequencies.

We also observe that when the forcing frequency is close to the first resonance, the two sliders are moving with (nearly) the same amplitude and with the same phase: we may give an interpretation for this by observing that in this frequency range the contribution of the first modal coordinate is predominant over the second one, so that the resulting motion of the system takes the “shape” of the first mode of vibration. On the other hand, at frequencies close to the second resonance the two sliders move with (nearly) the same amplitude but with opposite phase, which may conversely be explained by the fact that the contribution of the second mode of vibration becomes predominant in this different frequency region.

Thus, we recognise that there are specific frequency ranges where the behaviour of the system may be predicted (as a first approximation) by considering a single one out of the total modes of vibration, simplifying the description of the motion to that of a single-dof system.

The “boundary” between the two above described frequency ranges is represented in the Bode diagram of the left slider by a value of the forcing frequency where the amplitude of vibration goes down to zero while the phase increases by $+\pi$: this can be considered as a “dual” behaviour with respect to a resonance condition, where the amplitude of vibration becomes infinite and the phase decreases by $-\pi$, and for this reason this point is called an “antiresonance”.

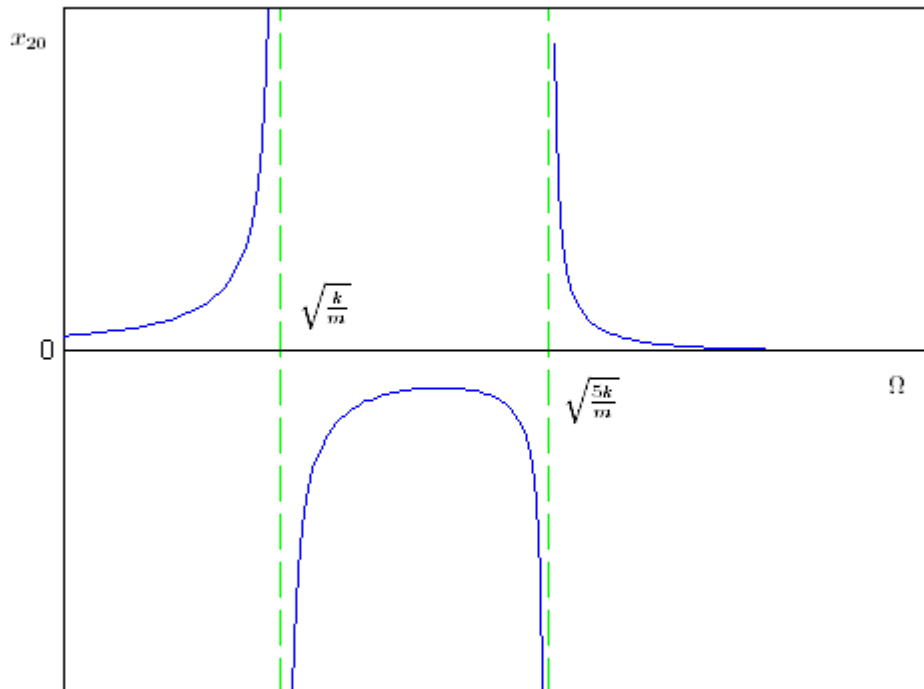
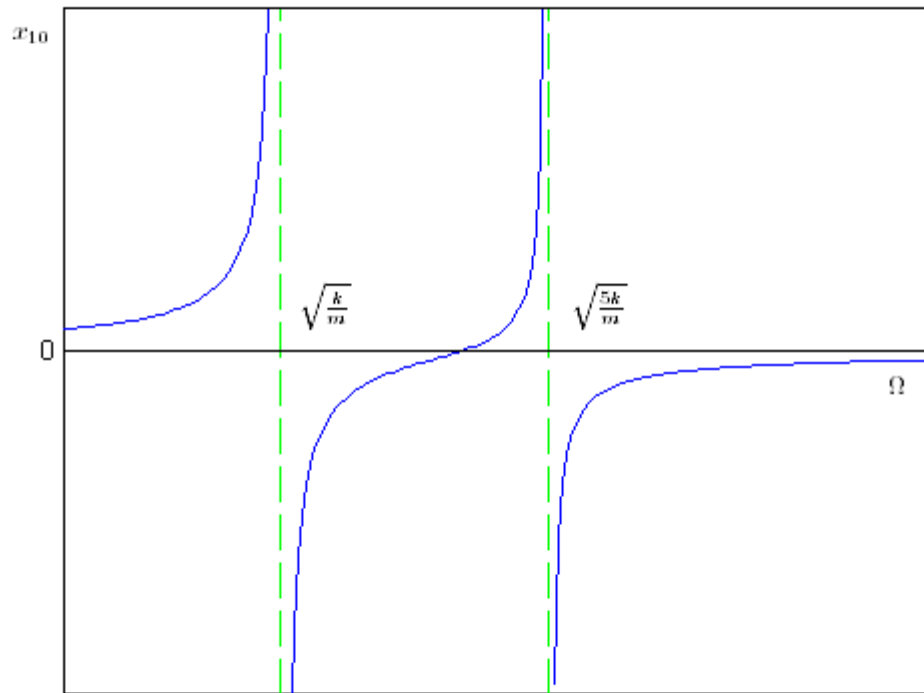


Figure 3.3: Diagrams showing the response of the “physical” coordinates x_1 and x_2 to harmonic forcing on the left slider, as function of the forcing frequency Ω - undamped case.

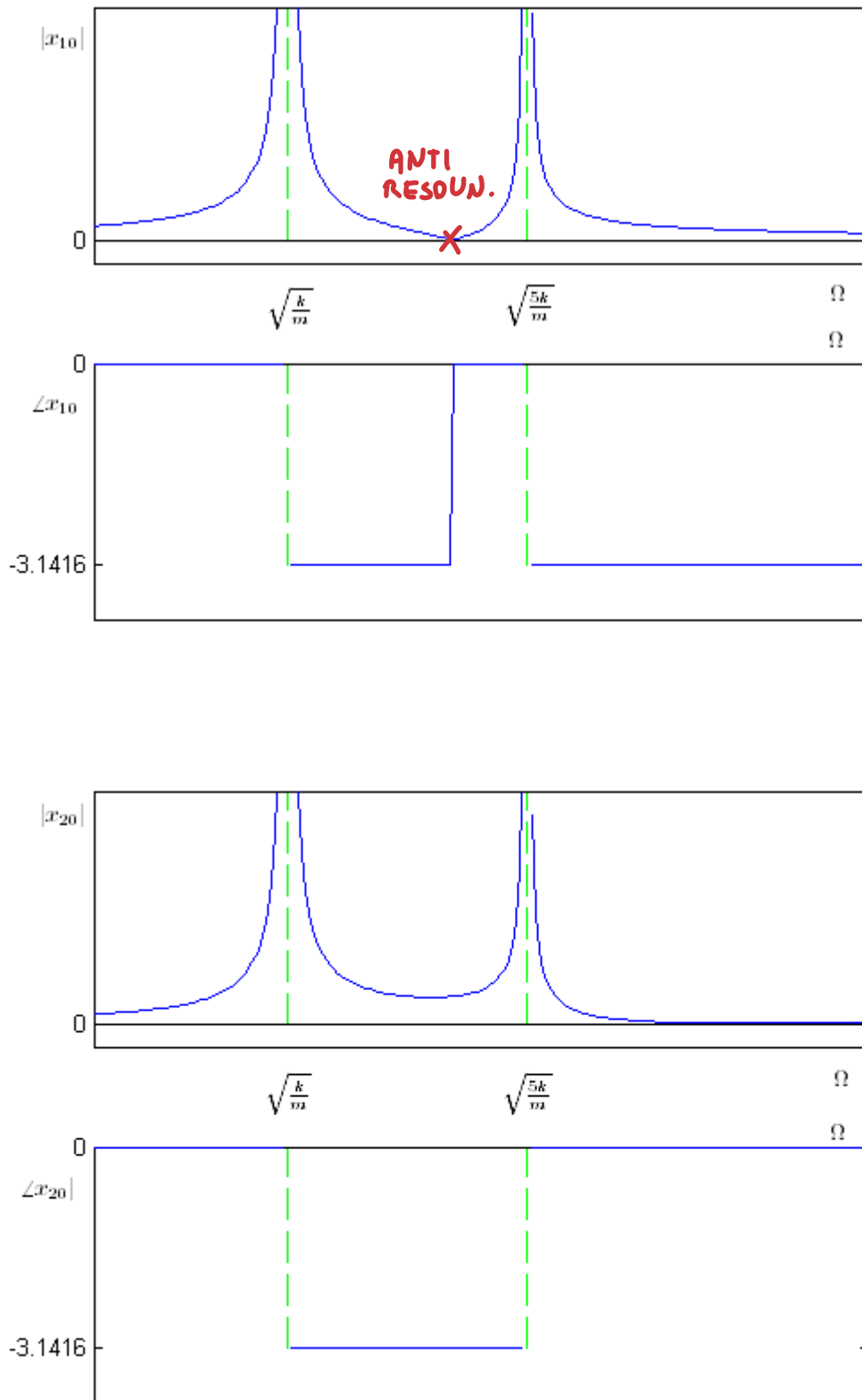


Figure 3.4: Bode diagrams of the displacements of the two sliders produced by a force applied on the left slider - undamped case..

In the case of a damped system ($c \neq 0$) the procedure to define the Bode diagrams of the FRFs remains the same, but the frequency response functions become complex, with the implication that the same mathematical operations outlined before can be used but have to be performed in complex algebra. However, in case of light damping the conclusion found by this study are similar to those obtained for the undamped system. In particular, the concepts of resonance, anti-resonance and predominance of the resonant modal coordinate apply also to a weakly damped system.

First of all, we note that for the damped system equations (3.6a,b) become:

$$\begin{cases} 2m\ddot{q}_1 + 2c\dot{q}_1 + 2kq_1 = f_0 \cos(\Omega t) & (3.11a) \\ 2m\ddot{q}_2 + 10c\dot{q}_2 + 10kq_2 = f_0 \cos(\Omega t) & (3.11b) \end{cases}$$

The steady state solution for these two equations is again represented by a harmonic vibration with the same frequency Ω as that of the forcing effect, like in the undamped case, but in this case a generic phase difference (normally different from 0 and $\pm\pi$) takes place between the input (force on the system) and the output (system's motion along the different coordinates). As was shown in lesson 2 for the more general case of a multi-dof system, this generic phase difference is well managed by the use of complex algebra to look for the solutions of equations (3.11a,b). Therefore, based on the observation that:

$$f_0 \cos(\Omega t) = \text{Re}(e^{i\Omega t})$$

we rewrite eq. (3.11a,b) in the complex form:

$$2m\ddot{q}_1 + 2c\dot{q}_1 + 2kq_1 = f_0 e^{i\Omega t} \quad (3.12a)$$

$$2m\ddot{q}_2 + 10c\dot{q}_2 + 10kq_2 = f_0 e^{i\Omega t} \quad (3.12b)$$

then we compute a solution for the above complex-valued equations of motion in the form:

$$q_1^*(t) = q_{01} e^{i\Omega t} ; \quad q_2^*(t) = q_{02} e^{i\Omega t} \quad (3.13)$$

q_{10} and q_{20} being complex-valued constant parameters, and finally we take the real part of that solution:

$$q_1(t) = \text{Re}(q_{01} e^{i\Omega t}) = |q_{01}| \cos(\Omega t + \angle q_{01}) ; \quad q_2(t) = \text{Re}(q_{02} e^{i\Omega t}) = |q_{02}| \cos(\Omega t + \angle q_{02}) \quad (3.14)$$

where $|q_{01}|$, $|q_{02}|$ denote the magnitudes and $\angle q_{01}$, $\angle q_{02}$ are the phases of the complex numbers q_{01} , q_{02} .

Introducing the complex solution (3.13) into eq. (3.12a,b) and after obvious simplification and re-arrangement we get:

$$q_{01} = \frac{f_0}{2k} \frac{1}{1 - \left(\frac{\Omega}{\omega_1}\right)^2 + 2i\left(\frac{\Omega}{\omega_1}\right)h_1} ; \quad q_{02} = \frac{f_0}{10k} \frac{1}{1 - \left(\frac{\Omega}{\omega_2}\right)^2 + 2i\left(\frac{\Omega}{\omega_2}\right)h_2} \quad (3.15)$$

where i stands for the imaginary unit and h_1 , h_2 are the damping factors of the 1-dof equations for the first and second mode respectively, which means eq. (3.11a) and (3.11b) respectively.

It is recalled that for a 1-dof whose equation of motion takes the form:

$$\bar{m}\ddot{q} + \bar{c}\dot{q} + \bar{k}q = f(t)$$

where $\bar{m}$, $\bar{c}$ and $\bar{k}$ are the generalised mass, damping and stiffness of the system, the damping factor is defined as the ratio between the generalised damping constant $\bar{c}$ and the “critical damping” of the system:

$$h = \frac{\bar{c}}{c_{cr}} ; \quad c_{cr} = 2\bar{m}\omega = 2\bar{m}\sqrt{\frac{k}{m}}$$

It is also recalled that the **critical damping is the minimum amount of damping that produces a non-oscillatory free motion of the system.**

From equations (3.15) it is **possible to draw the Bode diagrams of the two modal coordinates**, that are the plots showing the modulus and phase of q_{01} and q_{02} as function of Ω . These plots are shown in figure 3.5 for the case **$c=0.4 \text{ k}$** (it is left to the reader to verify that this leads to a damping ratio of 0.04 for the first mode and 0.089 for the second mode).

Finally, **to obtain the response of the system in terms of physical coordinates** (displacements of the sliders) **we use again eq. (3.9), which combined with (3.14) brings:**

$$\left\{ \begin{array}{l} x_1 = \text{Re}((q_{01} + q_{02})e^{i\Omega t}) = \text{Re}(x_{01}e^{i\Omega t}) = |x_{01}|\cos(\Omega t + \angle x_{01}) \\ x_2 = \text{Re}((q_{01} - q_{02})e^{i\Omega t}) = \text{Re}(x_{02}e^{i\Omega t}) = |x_{02}|\cos(\Omega t + \angle x_{02}) \end{array} \right. \quad (3.16)$$

where we see that the **amplitude and phase of the slider displacements may be obtained by taking the sum and difference of q_{01} and q_{02} , to be performed of course in complex algebra.**

The corresponding Bode diagrams of the slider displacements are shown in figure 3.6: it is noted that, **due to the light damping of the system** (the non-dimensional damping factor is much lower than one for both modal coordinates), **the Bode diagrams still show the same features as in the undamped case:** indeed **two resonance peaks are observed in both diagrams when the forcing frequency approaches one of the two undamped natural frequencies of the system** (actually, the maximum amplitude is not achieved when $\Omega=\omega_{1,2}$, but for very close values of frequency). In the neighbourhood of the resonance, the phase drops down by $-\pi$, like in the undamped case, except that **in presence of damping the phase variation is not a jump, but takes place continuously**. Finally, in the bode diagram of the left slider the **anti-resonance may still be detected as a value of forcing frequency for which the amplitude of vibration shows a sharp minimum** (the minimum valued being this time positive instead of zero), **which is accompanied by a phase variation of $+\pi$.**

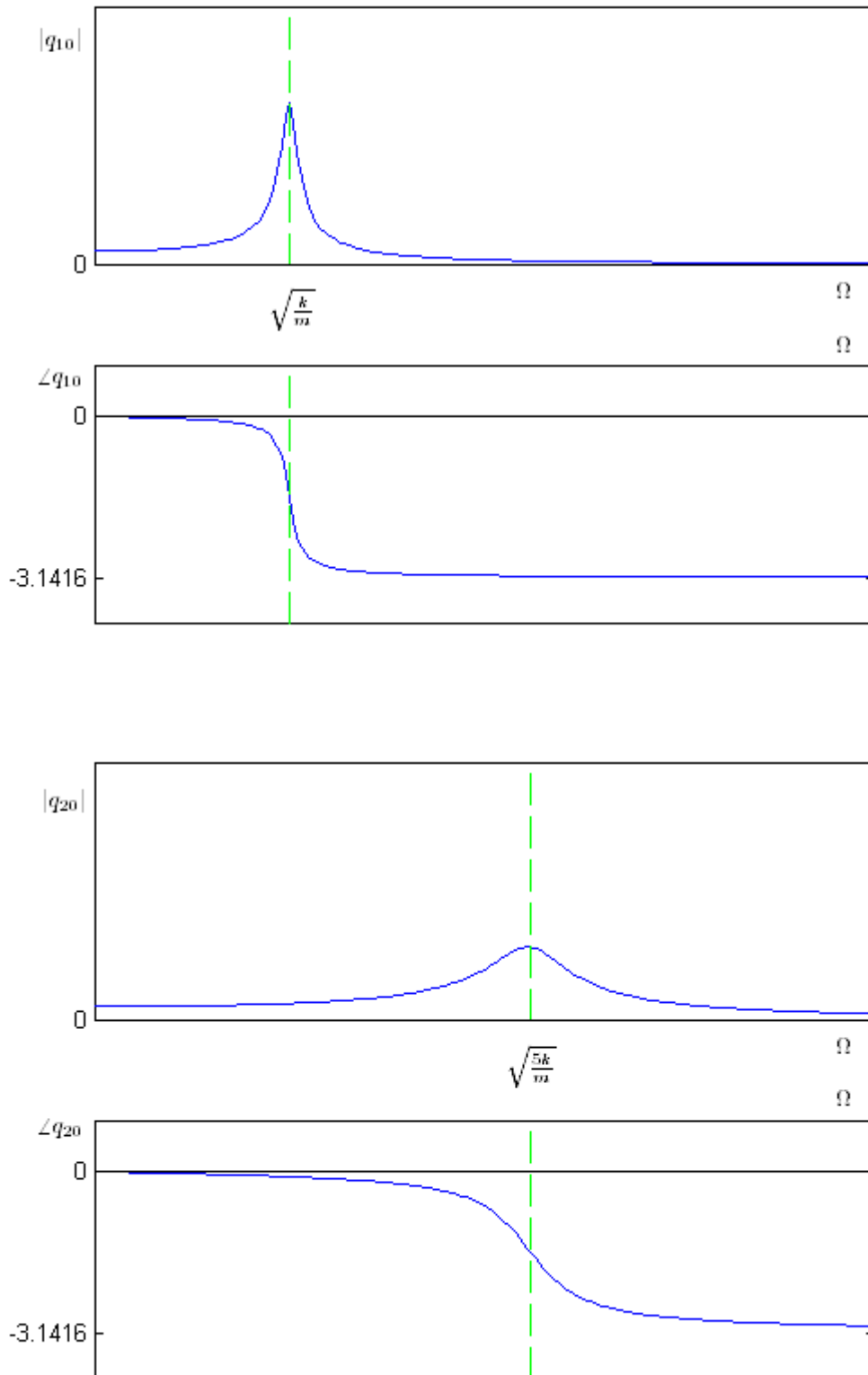


Figure 3.5: Bode diagrams of the displacement of the two modal coordinates produced by a force applied on the left slider – case with damping.

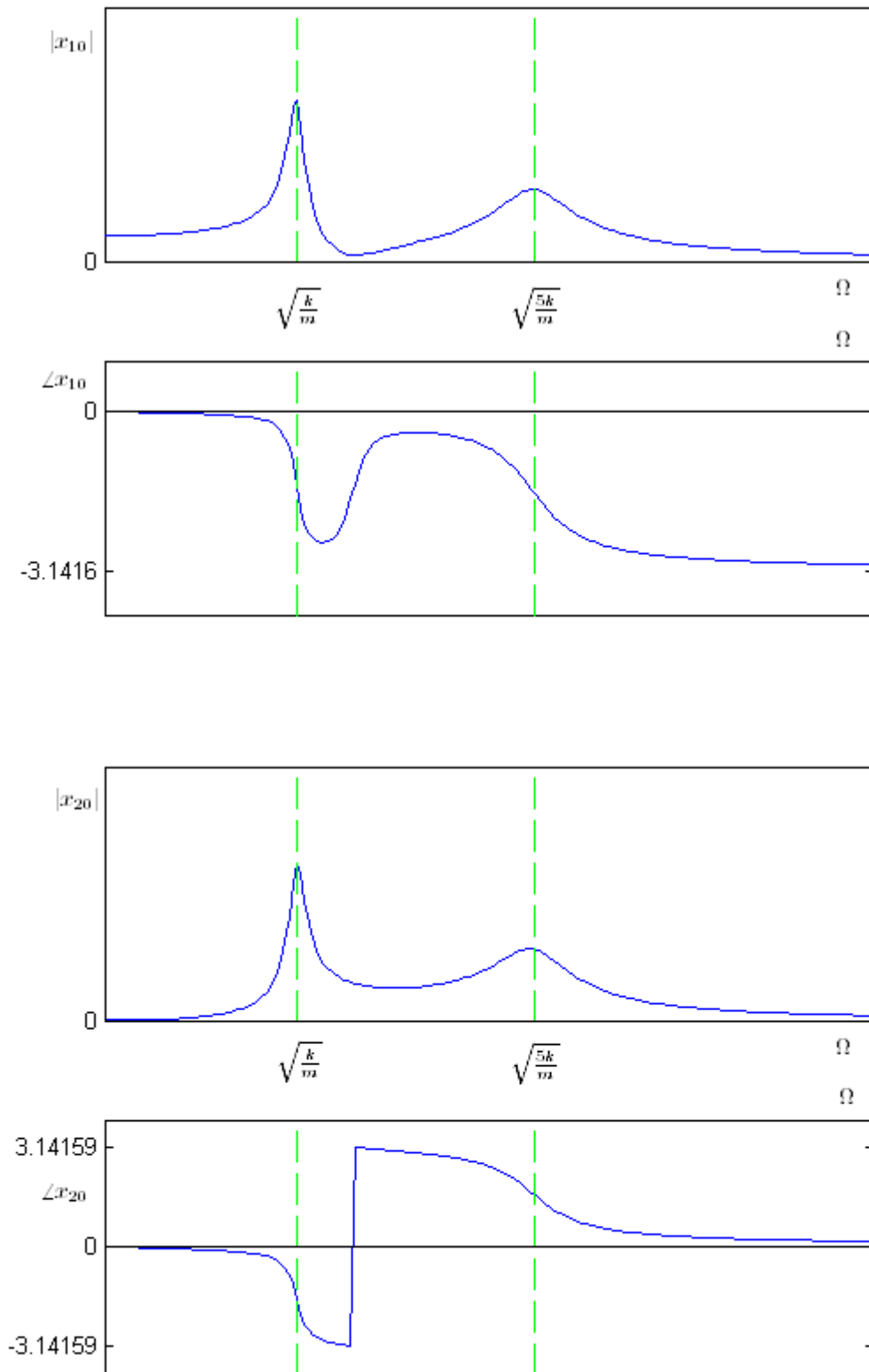


Figure 3.6: Bode diagrams of the displacement of the two sliders produced by a force applied on the left slider – case with damping.